

EE3621: Digital Signal Processing

Lecture 16: Linear-Phase FIR Realization & Frequency-Sampling Structure

(III B.Tech EEE)

September 9, 2026

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Design** Linear-Phase FIR structures with 50
 2. **Derive** the Frequency-Sampling filter realization equations from the IDFT formulation.
 3. **Construct** signal flow graphs for frequency-sampling FIR filters.
 4. **Evaluate** computational efficiency for narrowband filters with few non-zero DFT bins.
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2. MATHEMATICAL FOUNDATIONS

2.1 Linear-Phase Multiplier Reduction

For a symmetric FIR filter of length M ($h[n] = h[M - 1 - n]$):

- **Odd Length M (Order $N = M - 1$ Even):**

$$y[n] = \sum_{k=0}^{(M-3)/2} h[k](x[n-k] + x[n-M+1+k]) + h\left[\frac{M-1}{2}\right] x\left[n - \frac{M-1}{2}\right]$$

Total multipliers: $\frac{M+1}{2}$.

- **Even Length M (Order $N = M - 1$ Odd):**

$$y[n] = \sum_{k=0}^{M/2-1} h[k](x[n-k] + x[n-M+1+k])$$

Total multipliers: $\frac{M}{2}$.

2.2 Frequency-Sampling Structure Derivation

Express the FIR transfer function in terms of its M -point DFT samples $H[k]$:

$$h[n] = \frac{1}{M} \sum_{k=0}^{M-1} H[k] W_M^{-nk}$$

$$H(z) = \sum_{n=0}^{M-1} h[n] z^{-n} = \sum_{n=0}^{M-1} \left(\frac{1}{M} \sum_{k=0}^{M-1} H[k] W_M^{-nk} \right) z^{-n} = \frac{1}{M} \sum_{k=0}^{M-1} H[k] \sum_{n=0}^{M-1} (W_M^{-k} z^{-1})^n$$

Using the finite geometric sum $\sum_{n=0}^{M-1} u^n = \frac{1-u^M}{1-u}$, and since $(W_M^{-k} z^{-1})^M = W_M^{-kM} z^{-M} = 1 \cdot z^{-M} = z^{-M}$:

$$H(z) = \frac{1 - z^{-M}}{M} \sum_{k=0}^{M-1} \frac{H[k]}{1 - W_M^{-k} z^{-1}}$$

- **Comb Filter:** $H_{\text{comb}}(z) = \frac{1 - z^{-M}}{M}$ (FIR, zeros at $z = e^{j2\pi k/M}$).
- **Resonator Bank:** $H_k(z) = \frac{H[k]}{1 - W_M^{-k} z^{-1}}$ (All-pole, pole at $z = e^{j2\pi k/M}$).

Pictorial Representations

Symmetric Linear-Phase FIR folding structure ($M = 4$)

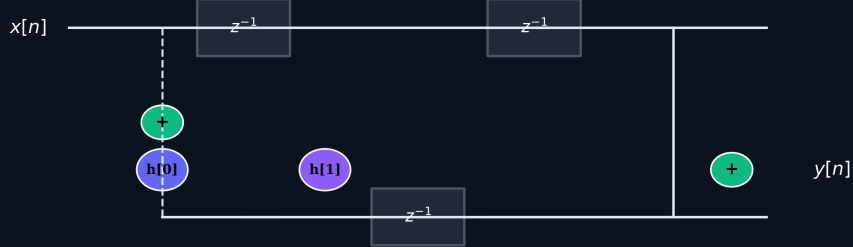


Figure 1: Linear Phase Fir

Frequency-Sampling FIR realization

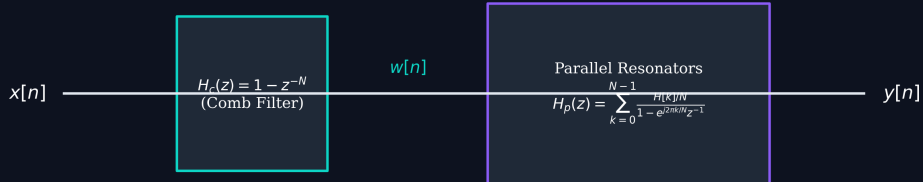


Figure 2: Frequency Sampling Structure

3. WORKED NUMERICAL EXAMPLES

Example 16.1: Linear-Phase FIR Structure

Problem: Draw the linear-phase direct form structure for an FIR filter with impulse response $h[n] = \{1, 2, 3, 2, 1\}$ ($M = 5$).

Solution:

- Length $M = 5$ (Odd).
- Symmetry: $h[0] = h[4] = 1$, $h[1] = h[3] = 2$, $h[2] = 3$.
- Output equation:

$$y[n] = 1 \cdot (x[n] + x[n-4]) + 2 \cdot (x[n-1] + x[n-3]) + 3 \cdot x[n-2]$$

- **Hardware Count:**

* Multipliers: $\frac{5+1}{2} = 3$ (compared to 5 in standard direct form). * Adders: 2 (pre-additions) + 2 (accumulation) = 4. * Delays: 4 registers z^{-1} .

4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) Derive the Frequency-Sampling structure for an FIR filter. Explain how pole-zero cancellation is achieved. *(8 Marks)* (b) Draw the linear-phase realization for an FIR filter of length $M = 6$ with $h[n] = \{-1, 2, 4, 4, 2, -1\}$. *(7 Marks)*

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**

* Complete IDFT substitution and geometric series derivation *(5 Marks)* * Explanation of comb zeros cancelling resonator poles on $|z| = 1$, and damping factor $r \lesssim 1$ for numerical stability *(3 Marks)*

- **Part (b):**

* $M = 6$ (Even length, symmetric): $y[n] = -1(x[n] + x[n-5]) + 2(x[n-1] + x[n-4]) + 4(x[n-2] + x[n-3])$ *(3 Marks)* * Multipliers required: $6/2 = 3$. * Neatly drawn SFG with 5 delay elements, 3 pre-adders, 3 multipliers, and accumulator bus *(4 Marks)*

5. PYTHON VERIFICATION SCRIPT

```
import numpy as np
```

```
\title{\textbf{EE3621: Digital Signal Processing}} \Large Verify linear-phase convolution
h = np.array([1, 2, 3, 2, 1])
x = np.array([1, 1, 1, 1, 1])
y = np.convolve(x, h)
print("Convolution Output:", y)
```